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Enhanced solid-state anaerobic digestion of corn stover by alkaline pretreatment

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ABSTRACT

Alkaline pretreatment was applied to enhance biogas production from corn stover through solid-state anaerobic digestion. Different NaOH loadings (1%, 2.5%, 5.0% and 7.5% (w/w)) were tested for solid-state pretreatment of corn stover. Lignin degradation during pretreatment increased from 9.1% to 46.2% when NaOH concentration increased from 1.0% to 7.5%. The NaOH-pretreated corn stover was digested using effluent of liquid anaerobic digestion as inoculum and nitrogen source. NaOH loading of 1% did not cause significant improvement on biogas yield. The highest biogas yield of 372.4 L/kg VS was obtained with 5% NaOH-pretreated corn stover, which was 37.0% higher than that of the untreated corn stover. However, a higher NaOH loading of 7.5% caused faster production of volatile fatty acids during the hydrolysis and acidogenesis stages, which inhibited the methanogenesis. Simultaneous NaOH treatment and anaerobic digestion did not significantly improve the biogas production ($P > 0.05$).

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1. Introduction

Lignocellulosic biomass, such as crop residues and forest waste, is mainly composed of cellulose, hemicelluloses and lignin. It has been well studied for ethanol production in recent years. The complex structure of native lignocellulosic biomass creates recalcitrance to enzymatic hydrolysis by cellulolytic microbes during anaerobic digestion. In addition, due to the low cellulolytic activity and low specific growth rate of cellulolytic microbes in anaerobic digestion, hydrolysis of native lignocellulosic biomass is usually slow (Noike et al., 1985). As a consequence, hydrolysis appears to be the most rate-limiting stage among the four stages (hydrolysis, acidogenesis, acetogenesis, and methanogenesis) of anaerobic digestion (Lu et al., 2007). Pretreatment of lignocellulosic biomass may represent a feasible solution to improve digestion efficiency and biogas production.

Alkaline pretreatment is one of the current leading pretreatment methods as it has shown several positive effects, including solubilization of lignin and neutralization of various acidic products degraded from the lignocellulosic complex (Hendriks and Zeeman, 2009; Pavlostathis and Gossett, 1985). Additionally, the presence of a small amount of residual alkali in the treated solids may be helpful for preventing the drop of pH during subsequent acidogenesis process (Pavlostathis and Gossett, 1985). Therefore, alkaline pretreatment is more effective and compatible with subsequent anaerobic digestion when compared to other pretreatment methods such as thermochemical pretreatment. Among the three kinds of alkali (NaOH, KOH, lime) tested for pretreatment of rice straw, NaOH was found to be most effective for lignin removal and biogas production (Yang et al., 2009). Zheng et al. (2009) reported that a 72.9% increase in biogas production was obtained from corn stover pretreated with 2% NaOH for 3 days. Although alkaline pretreatment, especially with NaOH, can lead to significant improvement on biogas production, toxicity of cations on methanogens was also reported (Feijoo et al., 1995). Compared to the conventional alkaline pretreatment, solid-state alkaline pretreatment only uses a limited amount of water to saturate the feedstock and there is no wastewater generated. Solid-state alkaline pretreatment of crop residues has been shown to be effective for improving biodegradability and biogas production. Pang et al. (2008) reported that corn stover pretreated with 6% NaOH at 80% moisture content and ambient temperature (20 ± 2 °C) for 3 weeks resulted in 48.5% more biogas yield over the untreated.

Solid-state anaerobic digestion (SS-AD) is generally used to process organic waste with high solid content (20–35%). It has been successfully implemented in Europe for processing the organic fraction of municipal solid waste (OF-MSW) (Bolzonella et al., 2003; Mata-Alvarez et al., 2000). Compared to liquid phase anaerobic digestion (0.5–15% solid loading), SS-AD is relatively stable and requires less energy input. However, the disadvantages of SS-AD include the requirements of large amounts of inoculum (up to 50%), much longer retention time (three times longer than liquid AD), and nitrogen supplementation when lignocellulosic biomass is used. In this study, an integrated anaerobic digestion

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Table 3
Change of VFA, pH and alkalinity during anaerobic digestion.

NaOH concentration (%)	pH_i[a]	pH_f[b]	$Alkalinity_i$ (g $CaCO_3$/kg)	$Alkalinity_f$ (g $CaCO_3$/kg)	VFA_f (g/kg)
Untreated	8.5 ± 0.0	8.5 ± 0.0	13.6 ± 0.7	11.9 ± 0.2	2.0 ± 0.1
1.0	8.5 ± 0.0	8.3 ± 0.0	14.3 ± 0.8	14.0 ± 0.1	2.6 ± 0.2
2.5	8.6 ± 0.0	8.5 ± 0.0	17.6 ± 0.9	15.3 ± 0.7	2.0 ± 0.7
5.0	8.9 ± 0.1	8.5 ± 0.1	21.8 ± 0.8	20.2 ± 1.3	2.4 ± 0.6
7.5[c]	9.5 ± 0.4	6.0 ± 0.3	26.4 ± 1.3	10.7 ± 0.5	22.1 ± 1.1

[a] *i* – initial values at the beginning of the digestion.
[b] *f* – final value after 40 days of digestion.
[c] *f* – final value after 20 days of digestion.

Corn stover pretreated with 7.5% NaOH caused failure of methane production. It was observed that corn stover pretreated with 7.5% NaOH resulted in a low daily biogas yield with low methane content and the digestion ceased on day 13 (Fig. 1a and c). A highly acidic environment (pH 6.0) caused by the accumulation of VFA inhibited the methanogenic bacteria when the 7.5% NaOH-pretreated corn stover was digested (Table 3). Similar results were reported by Penaud et al. (1999) who observed that methane production decreased when 5 g/L or higher NaOH loading was used. Although sodium ions were not found to inhibit the methane production in the study of Penaud et al. (1999), sodium concentrations of 0.25–0.4 M (10–16 g/L) were reported to impose sodium ion toxicity on methanogenic bacteria by Rinzema et al. (1988).

The cumulative biogas yields of corn stover pretreated with different NaOH loadings are presented in Fig. 1b. The highest cumulative biogas yield (372.4 L/kg VS) for 40-day digestion was obtained with 5.0% NaOH-pretreated corn stover, which was 37.0% higher than that of the untreated samples. The cumulative biogas yields for 40 days reached 266.8 and 275.9 L/kg VS for 1.0% and 2.5% NaOH-pretreated corn stover, respectively, which were not significantly different from the control (untreated corn stover). These results indicated that NaOH concentrations lower than 2.5% were not effective in improving the biogas production. On the other hand, 7.5% NaOH-pretreated corn stover inhibited the methane production.

Among the functional digesters, the methane contents increased gradually and reached about 60% on day 10 (Fig. 1c). There were variations of methane content among different treatments, but they were not significantly different. The methane contents ranged between 50% and 60% after day 10 for all the functional digesters.

Table 4
Degradation of substrates after 40-day anaerobic digestion.

NaOH concentration (%)	Degradation (%)			
	TS	VS	Cellulose	Hemicellulose
Untreated	21.8 ± 0.4	35.3 ± 2.3	44.5 ± 3.2	45.7 ± 3.8
1.0	21.2 ± 0.4	35.8 ± 0.3	43.2 ± 1.9	43.6 ± 3.2
2.5	22.2 ± 0.2	38.4 ± 2.0	44.1 ± 6.2	46.0 ± 3.3
5.0	25.4 ± 1.3	44.4 ± 0.8	65.1 ± 0.2	63.6 ± 2.5
7.5[a]	7.7 ± 0.4	13.7 ± 0.7	27.6 ± 2.5	61.3 ± 3.7

[a] Twenty days of digestion.

3.2.2. Degradation of corn stover

The efficiency of solid-state anaerobic digestion was evaluated in terms of TS and VS reduction as well as carbohydrate reduction. As shown in Table 4, VS reduction increased as NaOH loading increased from 1.0% to 5.0%. Pretreatment with 5.0% NaOH caused the highest VS reduction (44.4%) in SS-AD. No significant improvement on VS reduction was observed with 1.0% and 2.5% NaOH pretreatment. The TS reduction followed a similar pattern as that of the VS. Cellulose and hemicellulose reductions of 1.0–2.5% NaOH-pretreated corn stover during SS-AD were close to that of the untreated samples. The highest cellulose and hemicellulose reductions of 65.1% and 63.1%, respectively, were observed with 5% NaOH-pretreated corn stover, which is in agreement with the highest biogas production at this condition. The lowest cellulose reduction of 27.6% was observed with 7.5% NaOH-pretreated corn stover. The hemicellulose degradation of 7.5% NaOH-pretreated corn stover reached 61.3% which was higher than that of 1.0% and 2.5% NaOH-pretreated corn stover. This may be caused by the high alkali residue in the 7.5% NaOH-pretreated corn stover which continuously degraded the hemicellulose during the digestion process.

3.2.3. Variation of VFA, pH and alkalinity

Volatile fatty acids (mainly acetic acid, propionic acid, and butyric acid) produced during the acidogenic stages are vital to the anaerobic digestion process. The degradation of propionate and butyrate by syntrophic acetogenic bacteria (e.g. *Syntropher wolinii*, *syntrophomonas wolfei*) produces acetic acid that is subsequently degraded into methane and CO_2 by acetoclastic methanogens (Montero et al., 2008). The evolution of VFAs plays an important role in maintaining efficient anaerobic digestion as it strongly affects the pH value, alkalinity, and the activity of methanogens (Buyukkamaci and Filibeli, 2004; Moller et al., 2004). The irrevers-

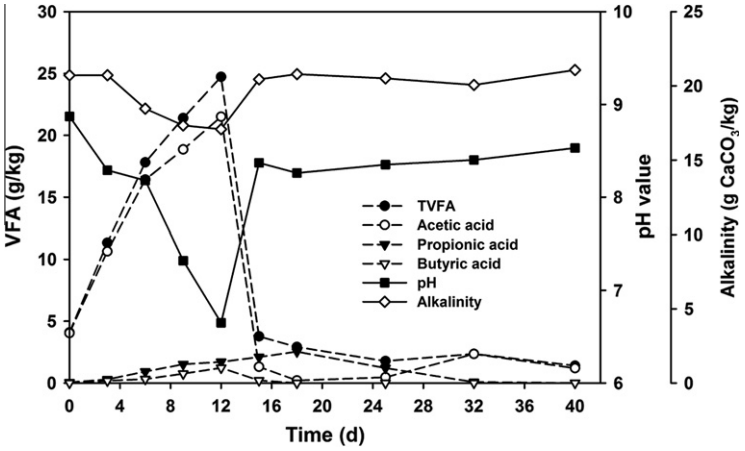

Fig. 2. Variation of VFA, pH and alkalinity during anaerobic digestion of 5.0% NaOH-pretreated corn stover.

ible acidification of the digestion, resulting from rapid hydrolysis and acidogenesis, is the major challenge of anaerobic digestion, as it can cause inhibition of methanogenesis or failure of the digestion (Veeken and Hamelers, 1999; Wang et al., 2009).

Variation of total short chain VFAs during anaerobic digestion of 5% NaOH-pretreated corn stover is presented in Fig. 2. Acetic acid was the dominant volatile fatty acids. There was no significant accumulation of propionic acid and butyric acid, probably because of the sufficient propionate- and butyric-degrading syntrophs in the inoculum which can rapidly convert propionic acid and butyric acid to acetic acid (Montero et al., 2008). The acetic acid increased rapidly after starting the test and reached a maximum of 21.5 g/kg on day 12. During this period, the acetic acid production rate was apparently higher than the acetic acid consumption rate. After 15 days, acetic acid concentration rapidly dropped to 1–2 g/kg and remained stable until the end of the digestion process. During this period, the digestion reached the stabilization stage (methanogenesis) with a balance between the production and consumption of acetic acid and a constant methane content of 50–60% was obtained.

The pH is an important parameter for proper operation and monitoring of the anaerobic digestion process. It strongly depends on VFA and buffering capacity. The pH value decreased with the increase in VFA production by acidogenic bacteria. The lowest pH value of 6.6 was observed on day 12, which corresponds to the maximum VFA production of 24.7 g/kg. Subsequently, pH values increased between 8.4 and 8.5 with a decrease in VFA concentration, indicating the further conversion of VFA to methane by methanogens. The alkalinity had a similar profile as pH, reaching the minimum level of 17.1 g $CaCO_3$/kg on day 12 and subsequently increased to 20.4–21.1 g $CaCO_3$/kg on day 12 (Fig. 2). The increase of alkalinity during the later stage could be mainly due to the consumption of VFA by methanogens, suggesting that the alkali (OH^-) remaining in the substrate solids after pretreatment did not increase buffering capacity of the digestion(Ahn et al., 2009; Kim et al., 2002).

The final VFA concentrations for each experimental condition tested are shown in Table 3. Acetic acid was the dominant component of VFA. For all the functional digesters, the VFA concentration fell between 2.0 and 2.4 g/kg and the final pH values were between 8.3 and 8.5, close to the initial values. In contrast, an accumulation of VFA (22.1 g/kg) was observed for 7.5% NaOH-pretreated corn stover at the end of the 20-day digestion, resulting in irreversible acidification of digestion with a pH decreasing from 9.5 to 6.0. The acetoclastic methanogens were reported to be severely inhibited by the accumulation of VFA and a decrease in pH (Powell and Archer, 1986). Separated acidogenic and methanogenic phases with leachate recirculation may provide buffering capacity and improve digestion performance (Myint and Nirmalakhandan, 2009; Veeken and Hamelers, 1999).

3.3. Simultaneous alkaline treatment and SS-AD

In order to simplify the process and reduce the capital cost, simultaneous alkali treatment and digestion (SATD) was also tested. Compared to the 24 h-pretreated corn stover, the highest peak daily biogas production of SATD was delayed from day 7 to day 9 for treatments with 1.0% and 2.5% NaOH addition. The 24-h pretreated corn stover can provide readily degradable carbohydrates to hydrolytic bacteria, which enhanced the biogas in the early stage. In SATD, corn stover needs to be depolymerized by alkali before hydrolysis and acidogenesis, resulting in no improvement on daily biogas production by alkali addition in the first several days. As shown in Fig. 3b, the 40-day accumulative biogas yield of corn stover with 1.0% NaOH addition was 278.9 L/kg VS, which was close to that of the untreated corn stover (275.9 L/kg VS). However, the biogas production at 2.5% NaOH loading in SATD between days 9 and 16 was higher than that of the untreated (Fig. 3a), which caused 9.0% increase in the cumulative biogas yield over the untreated.

SATD with 5% NaOH addition was inhibited at the early stage, resulting in a lower daily biogas production. The cumulative biogas production was 199.4 L/kg VS, which was 27% lower than that of the untreated. A high initial pH value (pH 9.42) caused by 5% NaOH addition might be the major reason causing the inhibition

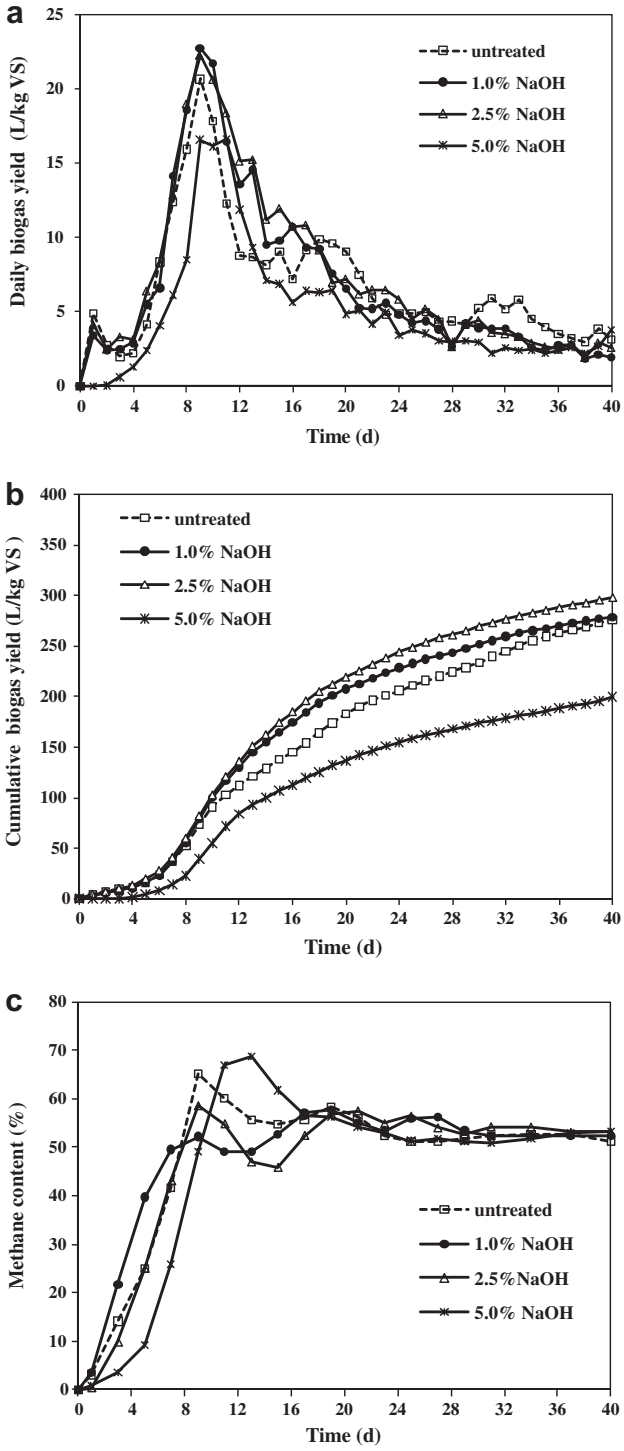

Fig. 3. Biogas production of simultaneous alkaline treatment and anaerobic digestion: (a) daily biogas yield, (b) cumulative biogas yield, and (c) methane content of biogas.

to the anaerobic microbial community, especially hydrolytic and acidogenic bacteria. As shown in Fig. 3c, the methane content of the biogas increased to 50–60% on day 7 for all the tests including control. Only slight changes in methane content were observed after day 16.

4. Conclusions

Solid-state alkali pretreatment for 24 h can effectively depolymerize corn stover with significant lignin degradation and limited cellulose loss. Solid-state anaerobic digestion of 5.0% NaOH-pretreated corn stover produced 37% more biogas when compared to that of the untreated. But a higher NaOH concentration (7.5%) caused inhibition of methanogenesis due to rapid hydrolysis and acidogenesis. The simultaneous alkaline treatment and digestion showed no significant improvement on biogas production.

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